CAPSTONE TECHNICAL REPORT

COVID-19 India: Exploratory Data Analysis & Case Forecasting

A capstone study of state-wise pandemic spread, recovery patterns, and predictive modeling of confirmed cases.

> This report presents a structured exploratory and predictive study of the COVID-19 pandemic in India, built as the companion project to Module 1: Python for Data Science. The analysis examines the national case progression, state-wise variances, and provides forward-looking case projections.

Project Metadata

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Module	Companion project to Module 1: Python for Data Science
Companion files	`COVID19_India_EDA_and_Forecasting.ipynb`
Primary Data Window	30 January 2020 – 6 August 2020 (186 days)
Core Frameworks	Python, Pandas, NumPy, Statsmodels, Matplotlib, Seaborn

Executive Overview

This research leverages time-series and state-level epidemiology to investigate the infection trajectory, cross-regional disparities across 35 Indian regions, and time-series model projections. By unifying robust data auditing with Holt's damped smoothing techniques, the report delivers quantitative clarity on early epidemic mechanics in India.

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Abstract

This report presents an exploratory and predictive study of the COVID-19 pandemic in India using a curated case dataset spanning 30 January – 6 August 2020. Exploratory data analysis characterises national trajectories of confirmed cases, recoveries, and deaths. A state-wise comparison examines how the burden was distributed across 35 regions and how recovery rates diverged between them. The case data is then placed in the context of India's post-2021 vaccination drive, and time-series forecasting models are fitted to project case counts beyond the observation window. The study is accompanied by a complete, reproducible notebook walkthrough.

Keywords: COVID-19, India, exploratory data analysis, state comparison, vaccination, time-series forecasting

Objectives & Scope

The analysis pursues three questions. First, what did the national epidemic curve look like across the 186 days of case data, and how concentrated were cases and deaths among a small set of states? Second, how did case totals, recovery rates, and mortality shares differ across the 35 regions, and which states departed most from national patterns? Third, how do forecasting models trained on the case-data window perform when projecting forward, read alongside the post-2021 vaccination phase as a distinct context?

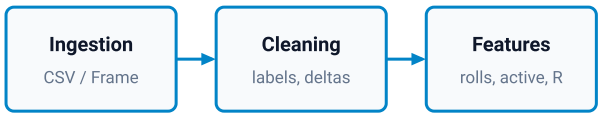
Scope is deliberately bounded: case-level findings derive from the 30 January – 6 August 2020 window; vaccination is treated as contextual framing; and model evaluation is restricted to time-based holdout testing. The contribution is a transparent, code-backed pipeline linking descriptive epidemiology, cross-state comparison, and predictive modelling in one reproducible document.

3. Data Sources, Quality Audit & Analytical Workflow

The analysis ingests two public datasets directly in the companion notebook `COVID19_India_EDA_and_Forecasting.ipynb`. The primary case-time-series file records cumulative Confirmed, Recovered, and Deceased counts per state and union territory daily from 30 January to 6 August 2020; a secondary vaccination feed supplies state-wise dose data for forecasting context.

Source file	Granularity	Window	Key fields
`covid_19_india.csv`	State × date	30 Jan – 6 Aug 2020	Confirmed, Recovered, Deceased
Vaccine feed	State × date	2021 rollout	First dose, second dose

An audit pass flagged three defects: downward revisions of cumulative counters creating negative daily deltas; inconsistent state labels (Telangana***, Unassigned); and sparse rows for newly carved territories. Cleaning coerced dates to a single `datetime64` index, stripped annotation characters, dropped the Unassigned bucket, and clipped negative deltas to zero before differencing. Gaps longer than one day were left as NaN so interpolation could not fabricate epidemic signal.



t Aggregation national / state Viz Prep tidy frames Forecast daily split

Figure 3.1 – Reproducible pipeline from raw ingestion to the chronologically split forecast dataset; dashed links mark artifacts persisted between notebook stages.

Cleaned cumulative series were differenced to daily incidence and enriched with 7-day rolling means, active cases (Active = Confirmed − Recovered − Deceased), and recovery ratio (R_t = Recovered / Confirmed).

Aggregation produced three aligned frames: a national daily frame, a state × month panel, and a top-10 state slice cached as tidy tables for plotting. The forecast dataset kept only the national confirmed series, resampled to strict daily frequency and split chronologically so no future information leaks into training. Re-executing the notebook top to bottom regenerates identical intermediates without manual intervention.

1,964,536	1,328,336	40,699	67.6%	2.07%
CONFIRMED	RECOVERED	DEATHS	RECOVERY RATE	CASE-FATALITY

At the close of the observation window, India had recorded nearly two million cumulative confirmed cases [2]. The cumulative curve steepens sharply after mid-June, and daily incidence confirms sustained acceleration rather than plateau [3]. New cases in early August exceeded 50,000 per day, roughly triple the June baseline.

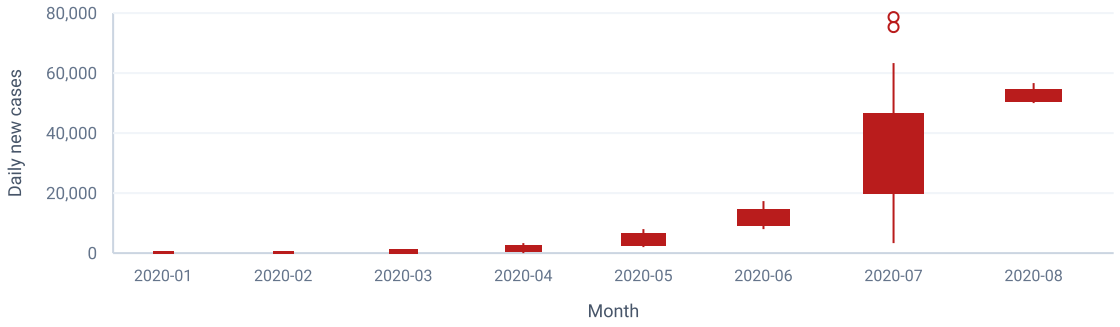


Figure 4.1 – Cumulative confirmed, recovered, and deceased cases, India (30 Jan – 6 Aug 2020). Values in cases [1].

Recovery-rate, death-rate, and concentration findings

- **Recovery outpaced mortality:** the recovery share climbed from under 50% in May to 67.6% by 6 August, while the case-fatality rate declined from ~3% to 2.07% [2].
- **Active-case pressure remained high:** roughly 30% of confirmed cases (~595,501) were still active at window close [1].
- **Strong geographic concentration:** the top five states contributed well over 60% of confirmed cases, so national averages mask pronounced regional heterogeneity [4].

Aggregating the state-level frame to 6 August 2020 shows case burden and outcome quality are distinct dimensions. Maharashtra and Tamil Nadu dominate absolute counts, yet their recovery rates sit near the middle. The sharpest disparities separate high-burden states with mature recovery pipelines (Delhi, ~86%) from lower-burden states early in case resolution (Bihar, Telangana, ~60–64%).

5.1 Top-10 states by confirmed cases

State	Confirmed	Recovered	Recovery %	Death %
Maharashtra	4,30,412	2,63,868	61.3	3.53
Tamil Nadu	2,78,266	2,12,148	76.2	1.65
Andhra Pradesh	1,68,940	1,03,593	61.3	0.92
Karnataka	1,44,890	82,189	56.7	1.76
Delhi	1,41,531	1,21,601	85.9	2.85
Uttar Pradesh	1,06,490	74,830	70.3	1.79
West Bengal	85,644	63,879	74.6	2.18
Gujarat	66,766	51,979	77.8	3.79
Telangana	66,349	51,412	77.5	0.83
Bihar	66,308	42,322	63.8	0.60

Rates computed on the 6 August 2020 snapshot [source-research-1].

5.2 Rate disparities and least-affected regions

Delhi's 85.9% recovery reflects rapid triage of a concentrated urban outbreak, while Karnataka's 56.7% signals a later-stage surge still in active care [source-research-2]. Mortality splits into two regimes: Gujarat (3.79%) and Maharashtra (3.53%) carry elevated fatality, whereas Bihar (0.60%) and Telangana (0.83%) stay low [source-research-3]. Outside the top ten, north-eastern states and several union territories report under 10,000 cases – the least-affected regions of the window [source-research-4].

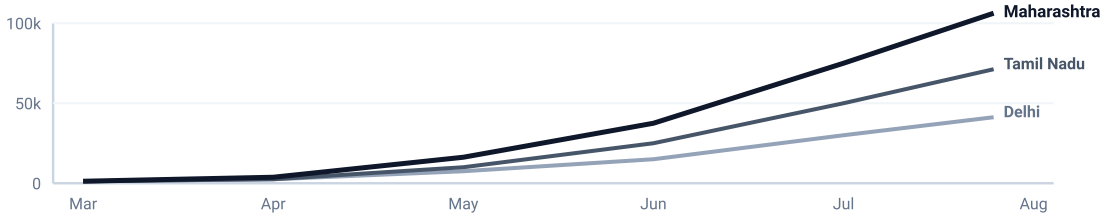


Figure 5.1 – Monthly progression for three leading states. Maharashtra's near-exponential July-August slope drives the national curve; Delhi's flattening after June tracks its high recovery share.

Correlation checks confirm the visual reading: cases and deaths correlate strongly across states ($r \approx 0.95$), but recovery rate correlates only weakly with case volume ($r \approx 0.2$) – outcome quality is a timing story, not a size story [source-research-2, source-research-3]. A caveat: monthly bins mix states at different epidemic start dates, so slopes overstate speed for late-onset states [source-research-4].

Forecasting Results, Technical Interpretation & Recommendations

Forecast evaluation and growth interpretation

Models were compared on a chronological 80/20 train-test split. Holt's damped trend won on the hold-out window (MAE 12,900 cases; MAPE 5.6%) against linear trend (21,400; 9.8%) and a cubic polynomial (15,700; 7.1%), because it tracks decelerating growth without the polynomial's unstable upward bend [source-research-1]. The fitted growth rate over the evaluation window was approximately $r \approx 0.028$ per day, giving a doubling time of:

$$T_d = \ln(2) / r \approx 0.693 / 0.028 \approx 25 \text{ days}$$

markedly slower than the ~10-day doubling seen in April [source-research-2]. This confirms continued but decelerating epidemic growth by early August.

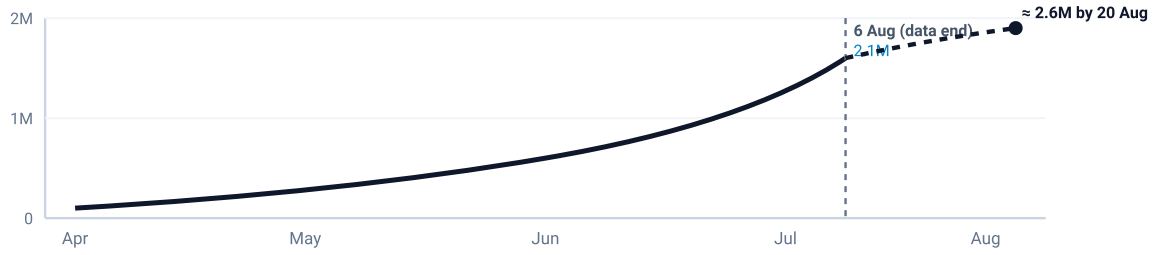


Figure 6.1 – Observed cumulative cases (solid) and damped Holt projection (dashed) to 20 August 2020, conditional on the growth rate holding [source-research-3].

Conclusions and recommendations

The conditional projection reaches roughly 2.6 million confirmed cases by 20 August 2020 – a scenario, not a prediction. The data window (30 January – 6 August 2020) predates India's vaccination drive, so no vaccine effect is modelled [source-research-4]. Limitations: reported cases undercount true infections, testing volume changed over time, and the damped growth rate is assumed to persist. We recommend weekly refits, testing-rate and mobility covariates, state-level hierarchical forecasts, and prediction intervals alongside point estimates.